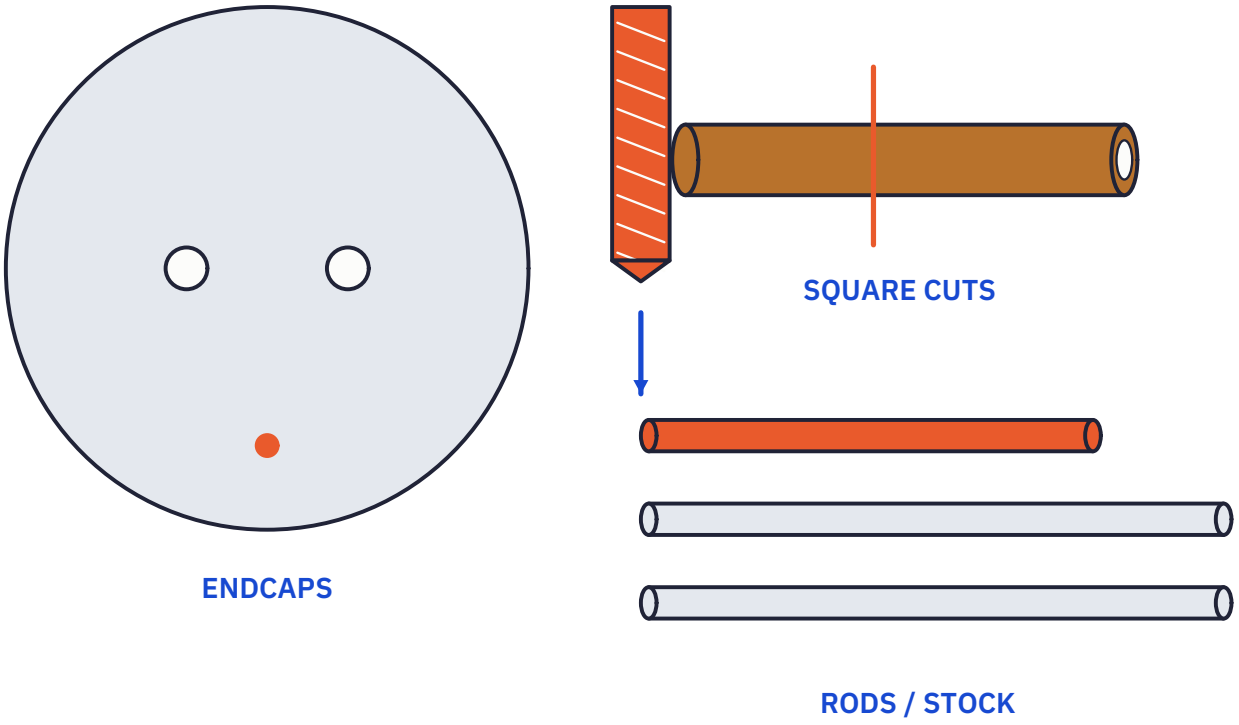
Drill & cut

A bench book for the pressure vessel, inlet jet, rods, tubing and harness stock.



A clean edge. A square cut. A part that fits.

Work one operation at a time. Coral shows the cutting action or the fresh edge; blue shows a gauge, clamp or measurement. Pictures explain the setup. Their size is not a cutting template.

BEFORE THE BATCH

Prove the setup and the first fit. A trial dimension or generated route is identified at the operation that uses it.

Choose the operation

Read the setup picture, make the cut, then check the part before the next operation.

03-08 **End plates + vessel tube**

Chamfer, NPT, blind registers, edge preparation

09-10 **Three float rods**

Blank lengths, square ends and register fit

11-17 **Water-inlet jet**

Fit, handling blank, clamping, drill, slice, section

18-19 **Copper + metal tubing**

Rotator shoe, clean tube and purge-link cuts

20-25 **Fluid tubing + coverings**

Square cuts, faucet, internal routes, elbow tubes, braid

26-27 **Wire + weld cleanup**

Ribbon stock, branches and attached filler wire

28 **Keep the source at hand**

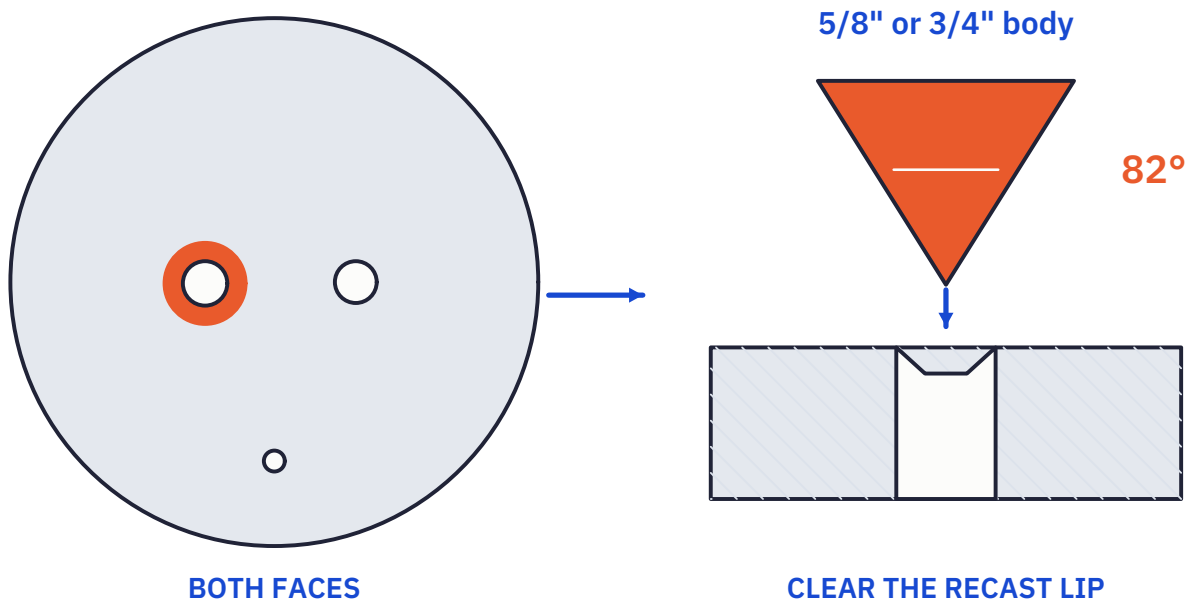
Current procedures and companion guides

AT A POWERED CUT

Read the WEN manual. Clamp work and vise to the table, remove the chuck key, use eye/hearing protection, keep hair/clothing clear and wear no gloves at a running press or saw.

Chamfer the port holes

Clear the laser-cut lip before the taper tap touches either plate.



Four port holes per carbonator. Break the lip on both faces of every 7/16-inch pilot.

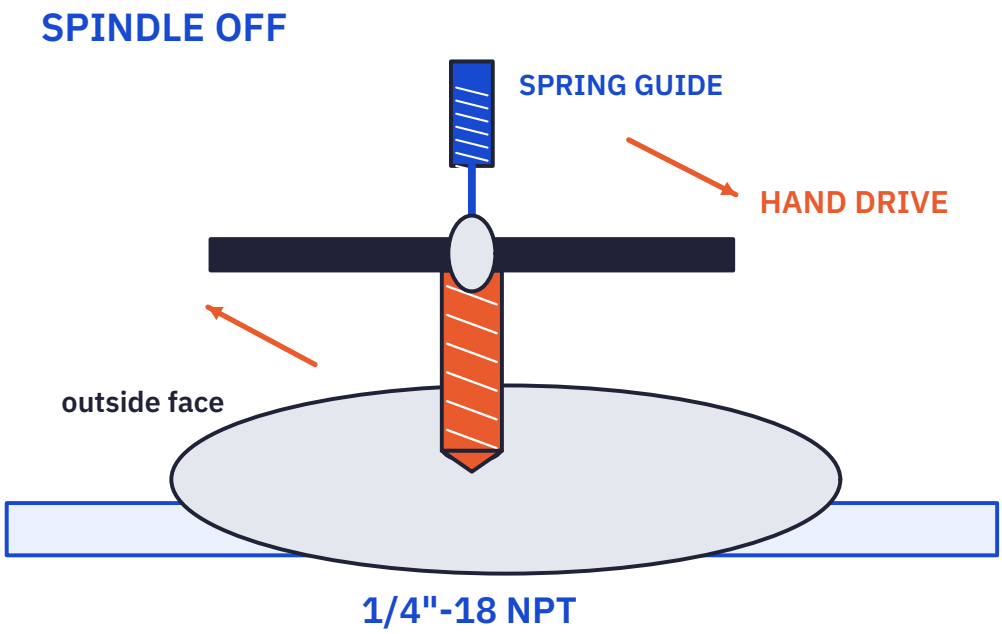
1. Clamp the plate and backer securely. Fit the JNB Pro 82° countersink with the press unplugged; use its **5/8-inch or 3/4-inch body**.
2. Apply Tap Magic EP-Xtra. Run slowly and touch in lightly; crowding the five-flute cutter makes it chatter.
3. Turn the plate over and clear the other face. Remove chips with the spindle stopped, then inspect all four hole edges.

CHECK THE LIP

The recast lip is gone and the tap has a clean square start. This step does not enlarge the pilot to make a fitting pass.

Hand-tap from outside

The drill press holds the spring guide square. Your tap wrench supplies the drive.



1/4-inch-18 NPT. Two holes in each plate, entered from that plate's outside face.

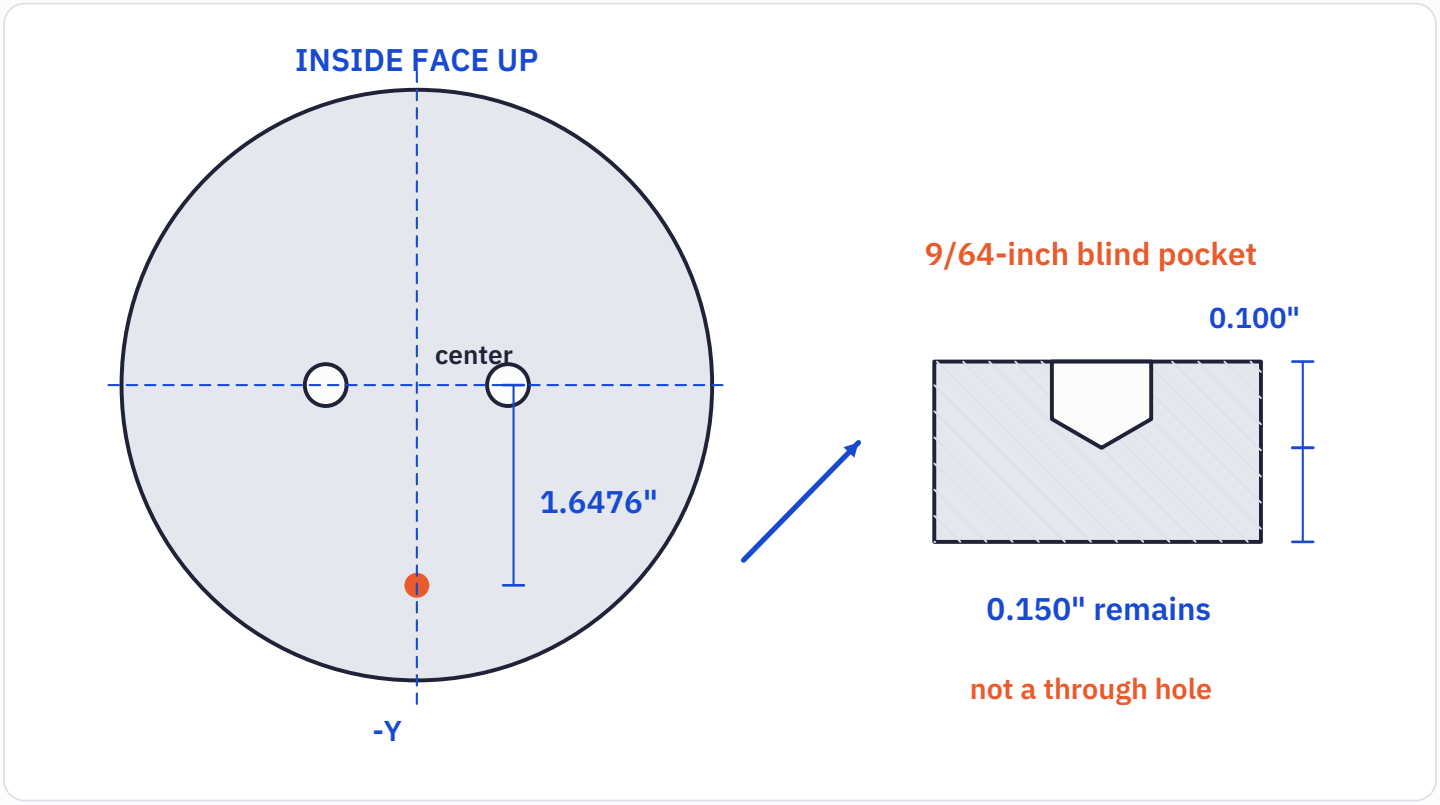
1. Mark the outside face and secure the disc. With the spindle **off**, align the Brown & Sharpe spring guide with the tap axis.
2. Lubricate the LingGan M35 taper tap with Tap Magic. Turn the wrench by hand; clear chips and re-lubricate as cutting requires.
3. Check with the actual elbow. Target **4.5 turns of engagement**, snug-firm with **2-3 threads showing**; verify the final clock on a trial before the batch.

FIRST FIT BEFORE FORTY PORTS

The production tapping fixture and repeatable engagement still need a recorded trial. Tap depth is decided by the fitting check, not a turns-only shortcut.

Locate the blind register

Both identical plates get the same pocket in their inside face.



The center is (0, -1.6476 in), 41.849 mm from the disc center, perpendicular to the two-port axis.

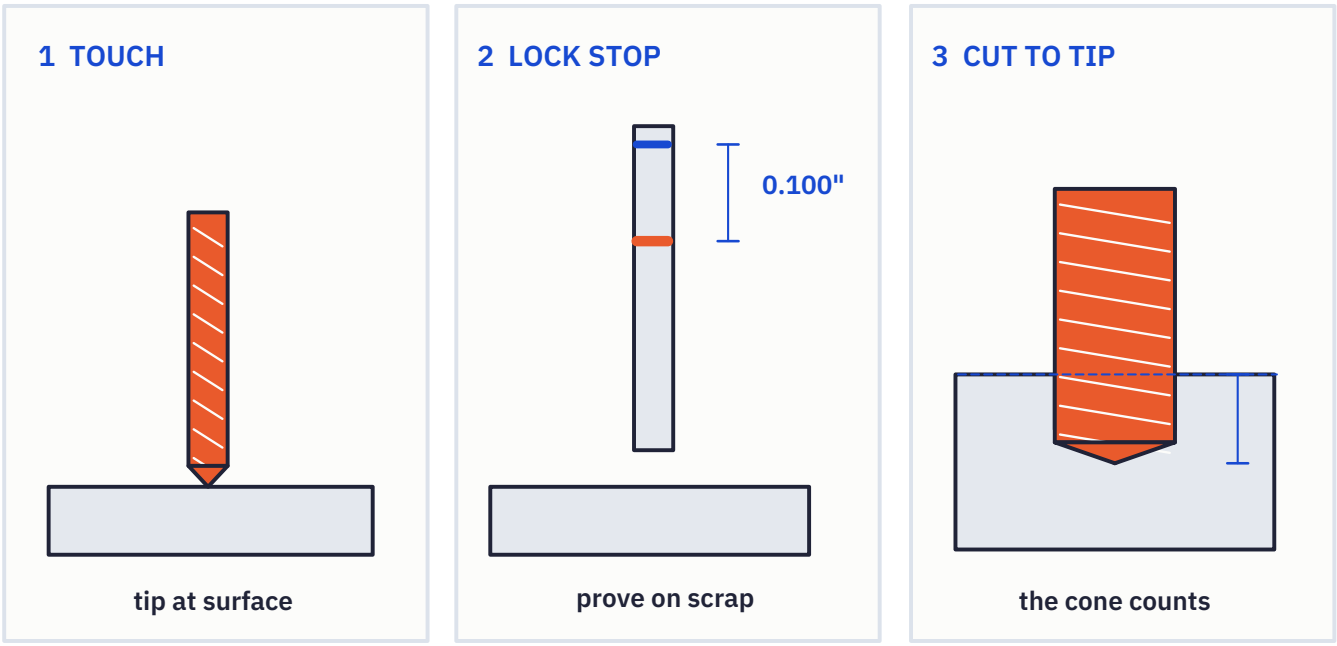
1. Put the **inside face up**. Find the disc center and the axis through the two port centers; lay out the perpendicular -Y axis.
2. Mark the register center **1.6476 inches / 41.849 mm** from the disc center, clear of both ports.
3. Align the stopped **9/64-inch M35, 135° split-point** drill to the mark, then clamp the disc and backer. Recheck the tip after tightening.

SAME POCKET, TWO JOBS

The bottom pocket seats the rod for its tack; the top pocket captures its tip. The pocket is blind because the end plate is the pressure boundary.

Prove the depth stop

0.100 inch means the drill-point tip, including the cone at the bottom.



A 0.100-inch tip depth in a 0.250-inch plate leaves 0.150 inch intact.

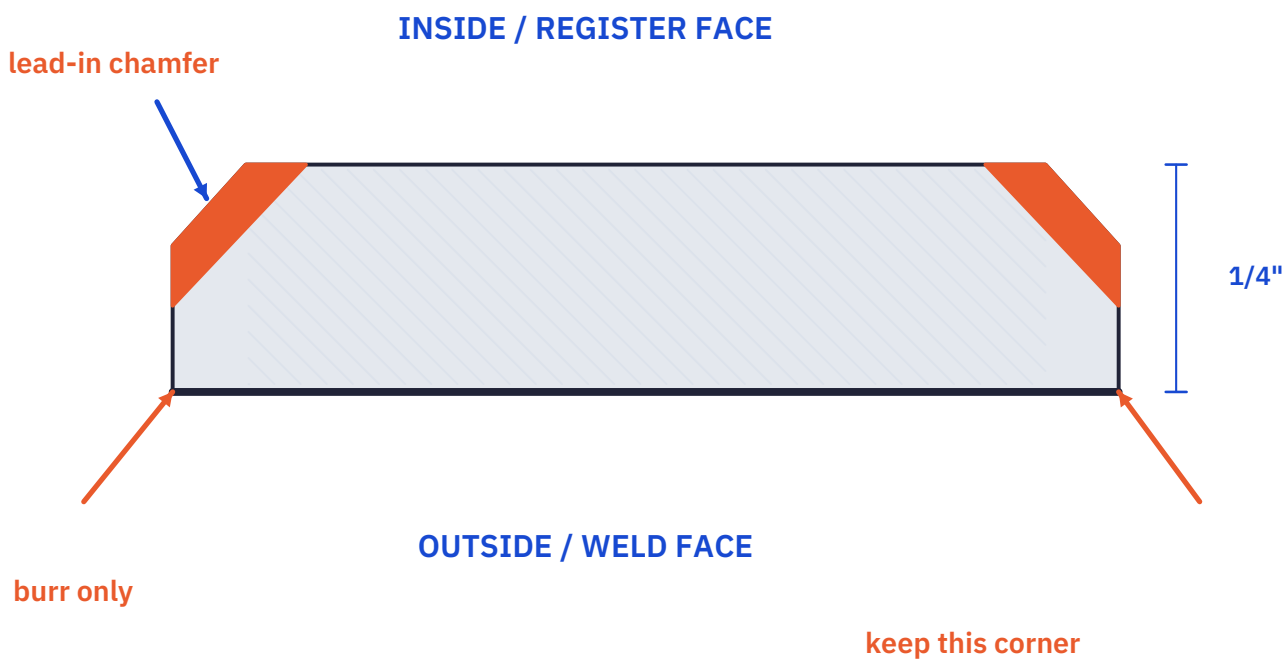
1. With the press stopped, touch the drill point to the inside surface. Set and lock the stop for **0.100 inch of tip travel**.
2. Prove the depth on a scrap disc before a real end plate. Use the slowest WEN setting, **about 740 rpm**, with Tap Magic.
3. Drill each clamped plate to the proved stop, clearing chips and renewing fluid. Remove the burr without increasing the pocket depth.

KEEP THE BACK FACE INTACT

Stop if the setup moves or the depth is uncertain. A through-hole rejects this pressure-boundary plate; it is not a weld repair step.

Two faces, two edges

The plate needs a lead-in inside and a sharp fillet root outside.



Section through the perimeter. Coral highlights the inside lead-in; the outside corner remains crisp.

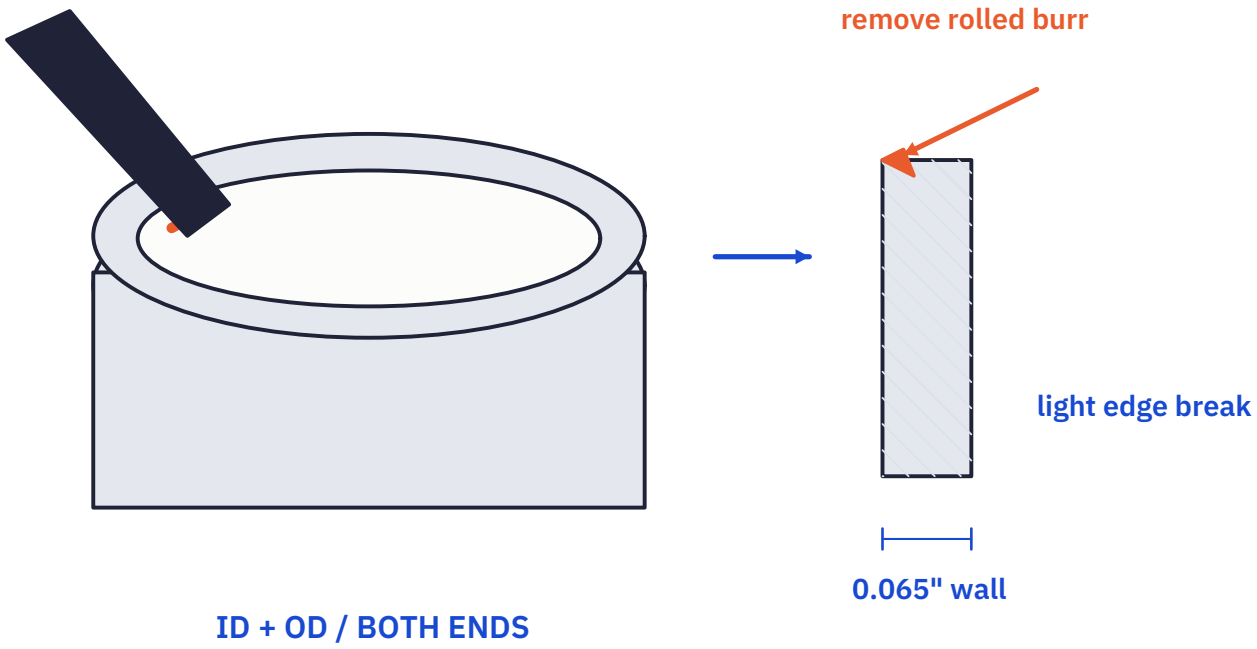
1. Keep track of the register face. Run the Noga NG8150 around the **inside-face OD edge** to make the insertion lead-in.
2. On the **outside-face OD edge**, remove the burr only. Keep the corner where the outer face meets the tube bore crisp.
3. Inspect and clean both edges. Keep stainless-only abrasives separate from anything used on carbon steel.

PRESERVE THE WELD ROOT

A broad outside chamfer opens the gap the laser must bridge. Coarse grinding can round this corner even when the plate still looks tidy.

Deburr the vessel tube

Prepare both loose tube ends while you can turn the tube freely.



The vessel is 5-inch OD × 0.065-inch wall × 6-inch cut length. The closure fillet sits 1/4 inch below the cut rim.

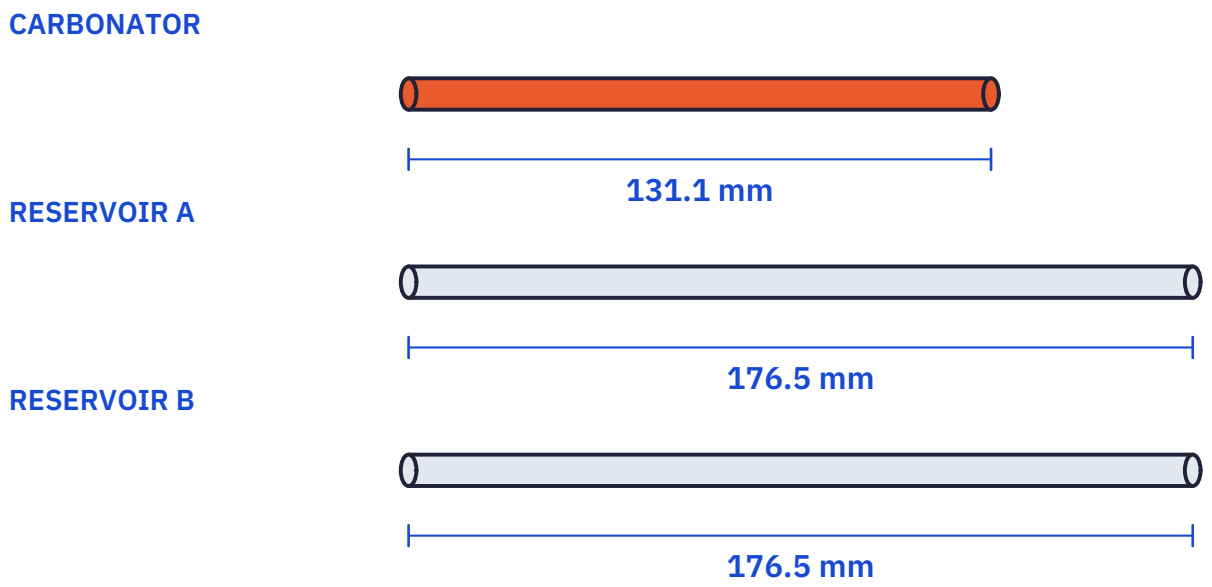
1. Check the received cut ends. Use the Noga to remove rolled saw burrs from **ID and OD at both ends**; keep the edge break light.
2. Check that the plate slips into the bore and seats at its 1/4-inch recess. A rim burr must not catch and hold it above the seat.
3. Clean the tube's bore band below each end and the plate's outside face. Keep the actual rim and end-cap seat square for the rotator checks.

SQUARENESS IS A FIT CHECK

The rotator's radial and face runout gates still apply. If a received end needs squaring, support the tube and check the corrected length and seat before welding.

Cut the three float rods

One carbonator blank and one rod for each flavor reservoir.



Square-cut and deburr 1/8-inch 316L stock. The carbonator value is a starting blank; the reservoir value already includes end clearance.

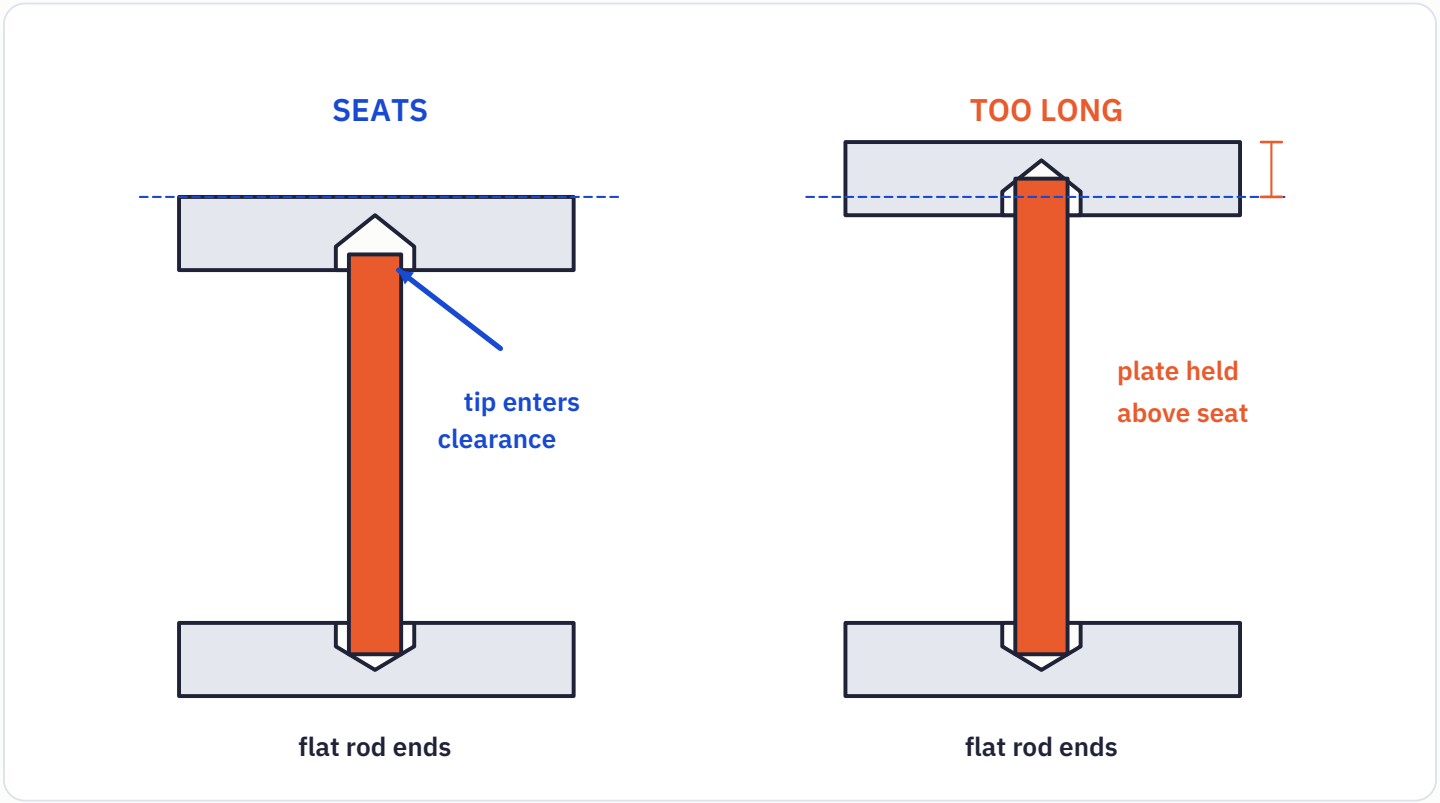
1. Mark one **carbonator blank** and two **reservoir rods** to the lengths above. Identify each before the cut.
2. Set a square cut at the BA4555. Secure the retained rod in the vise with support that prevents rocking; keep the cutoff free of the blade and stop.
3. Cut, deburr both flat ends and measure each rod. The vessel rod is hand-fitted at the next step; keep the two reservoir rods paired with their bodies and caps.

NO CAP HELD OPEN

Rod capture locates a cap; it never supplies its closing stop. Check both end registers and the seated cap rather than accepting the cut length alone.

Hand-fit the carbonator rod

Conical pocket bottoms make the actual seat-to-seat span shorter than the tip-depth arithmetic.



The 131.1 mm blank is hand-fitted to the actual two plates before the bottom tack.

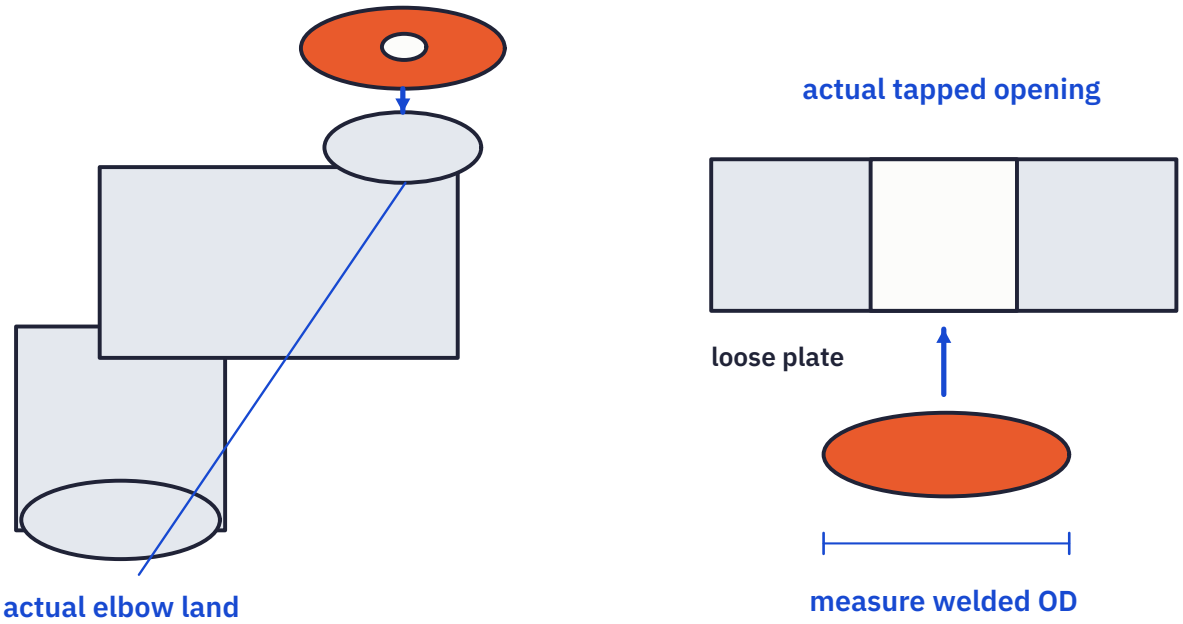
1. Seat the flat-ended rod in the bottom pocket. Check the real upper plate at its specified **1/4-inch recess** before committing the tack.
2. Shorten and deburr the rod as needed so its tip enters the top pocket while the plate reaches its full seat. Leave clearance at the top.
3. Repeat the seating check with the actual float and parts. The reservoir rods are captured by printed bosses and are **not welded**.

A POCKET DOES NOT SET THE RECESS

Positive tip engagement and a fully seated plate must both be present. Open the top pocket to 5/32 inch only if actual binding calls for it; keep it blind.

Measure before making jets

The nominal 9.5 mm × 2 mm cap is a fabrication-trial candidate.



Use the actual rod, elbow and a loose tapped plate. CAD elbow dimensions do not establish this fit.

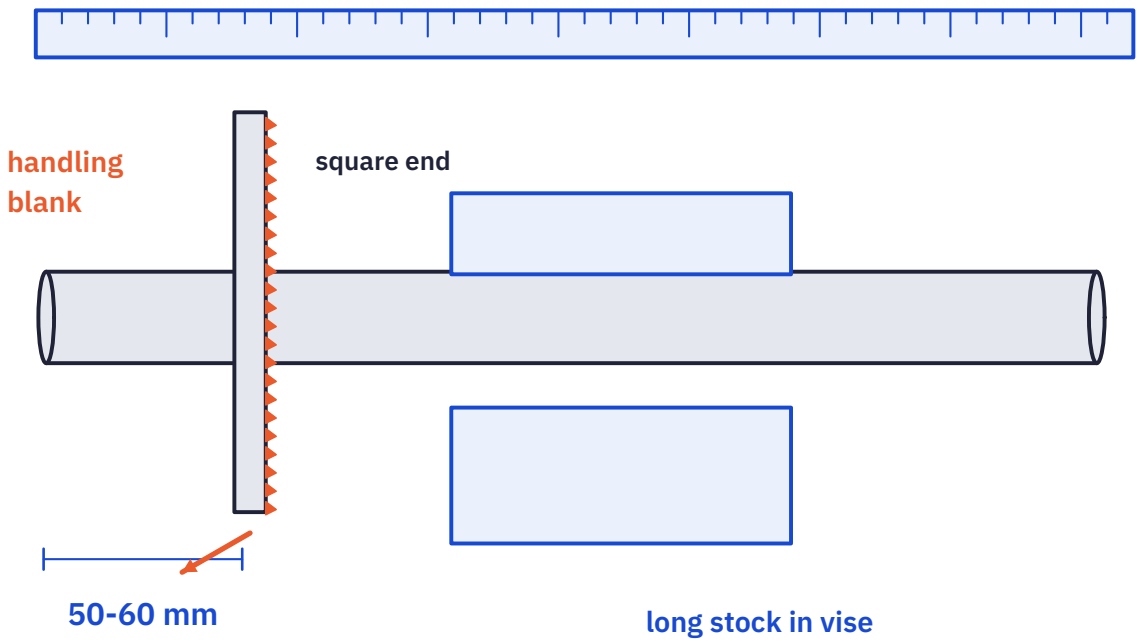
1. Measure rod OD, elbow tip OD, bore and flat annular land. Check that the cap covers the bore and seats flat with a root that can be fused.
2. Measure the tapped plate's clear opening and the finished welded OD. Check made-up elbow depth, clock and headspace projection.
3. Record the accepted fit before cutting a batch. A cap that will not clear the plate returns to design; it does not justify a deeper tap.

TRIAL / UNQUALIFIED

The cap fit, drilled/cut repeatability and weld root have no accepted measurements recorded. Use a spare elbow and retain the trial records.

Cut a handling blank

Keep the full 400 mm jet rod off the upright drill-press setup.



A 50-60 mm handling blank gives the two-piece fixture a manageable first setup.

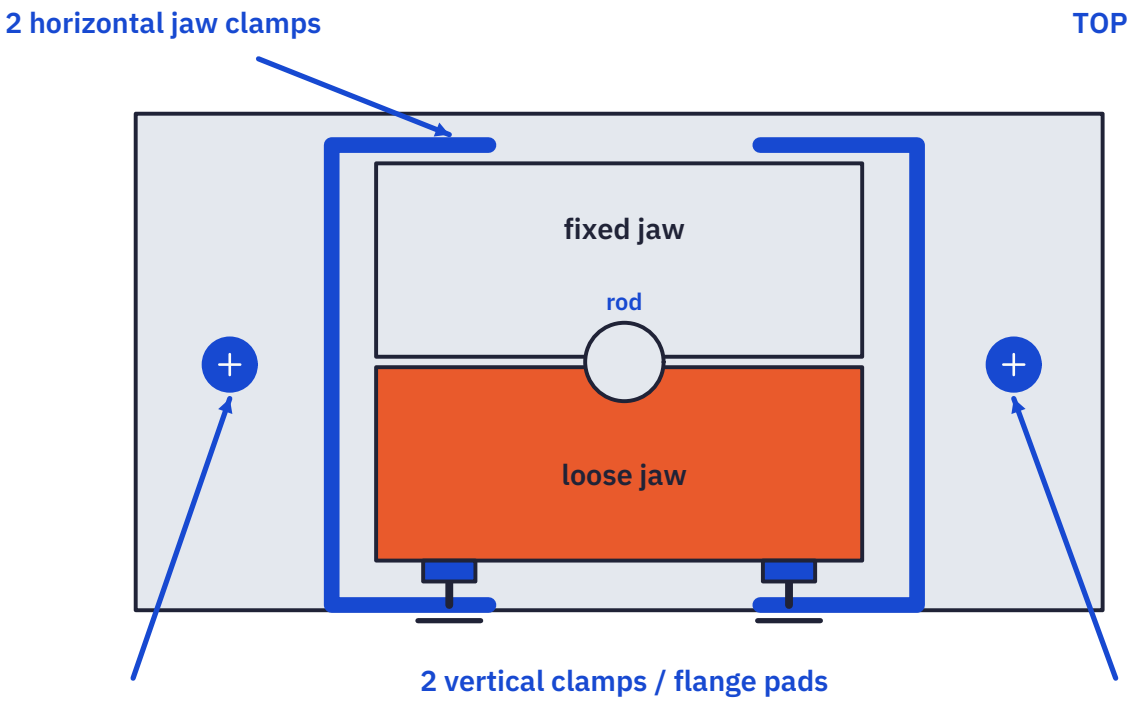
1. Mark **50-60 mm** on the nominal 9.5 mm 316 rod. Set the BA4555 cut square and clamp the long retained stock in its vise.
2. Use the 24 TPI M42 blade trial at low speed. Support the short blank's landing and keep the cutoff free to fall.
3. Deburr both ends flat. Clear chips from the fixture floor before the blank goes into the drill jaws.

KEEP ENOUGH ROD TO GRIP

Retire this handling blank before it falls below 40 mm, or earlier if chuck clearance requires it. Keep the short remainder as coupon stock.

Clamp the jet blank

Two clamps close the split jaws; two more secure the base and backer to the table.



Top schematic. The rod seats on a solid floor. Check real clamp frames and the stopped quill through the full drilling stroke.

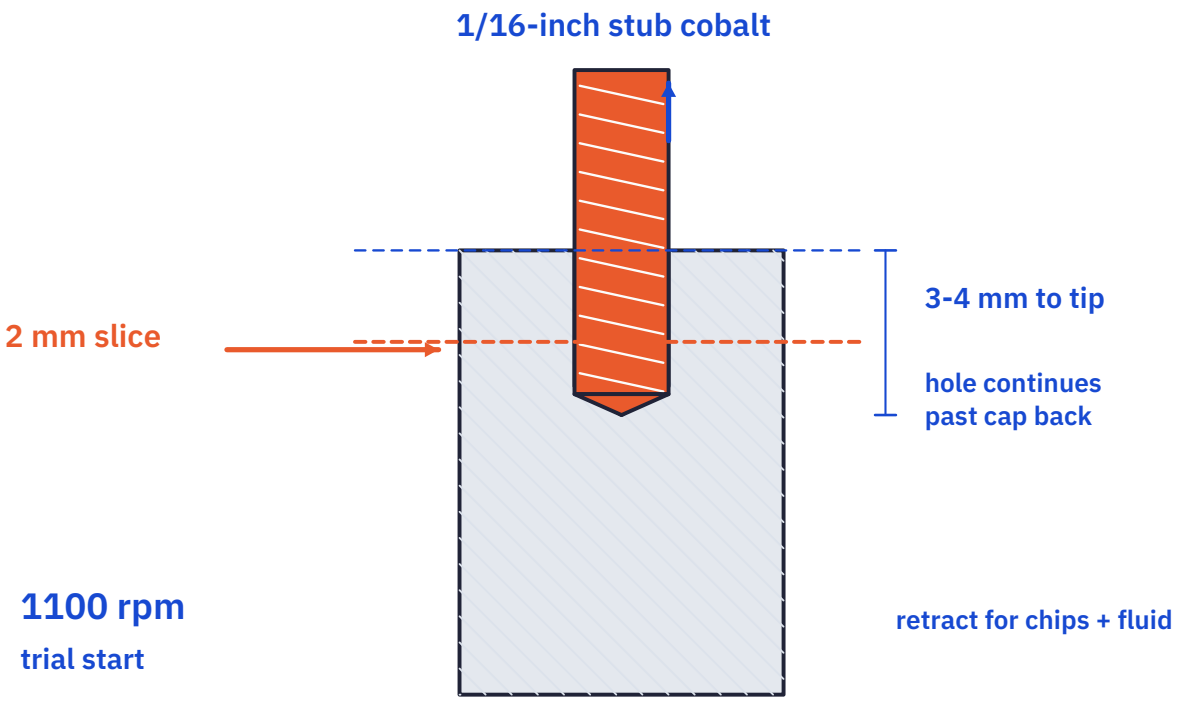
1. Seat the flat blank on the solid floor. Tighten the two **horizontal jaw clamps** evenly until it cannot turn, rock or lift; the jaw split stays visible.
2. Align the stopped drill after tightening the jaws, then secure the two **vertical table clamps** against a solid table edge, trapping the base and backer.
3. Turn the chuck by hand and lower the stopped quill through the intended stroke. Prove grip and clearance on scrap before caps.

PRINTED GRIP NEEDS A TRIAL

The jaw fixture is CAD-checked, physically untested. Stop at stock or fixture motion, jaw opening, rubbing or drill wander; do not solve poor grip with drilling force.

Drill the jet first

The cap stays on its handling blank until its short axial passage is drilled.



The drill's full-diameter hole must continue past the proposed 2 mm slice. Do not drill the whole handling blank.

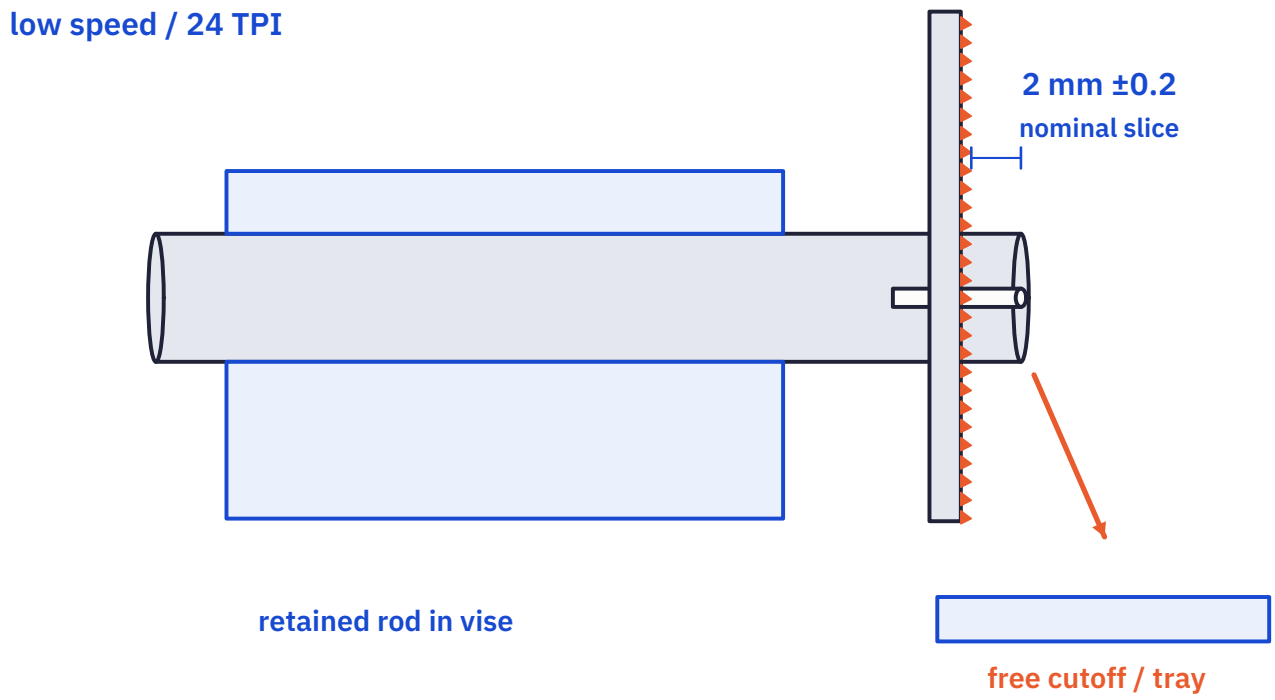
1. Align the **1/16-inch stub cobalt drill** to the rod axis. Verify concentric chuck grip and all four clamps before powering the press.
2. Begin at the WEN's **1100 rpm trial setting** with Tap Magic. Feed to make chips; retract to clear them and renew fluid.
3. Drill about **3-4 mm deep**. Stop for squealing, heat, stalled cutting or wandering; correct the setup before continuing.

A STARTING SETTING

1100 rpm is not an accepted production recipe. Record the setting and the cut that actually work with this stock, drill and clamping setup.

Saw the 2 mm cap slice

Take the drilled blank out of the printed jaws and hold the retained rod in the saw vise.



Put the drilled end on the cutoff side. Do not trap a thin slice between the moving blade and a stop.

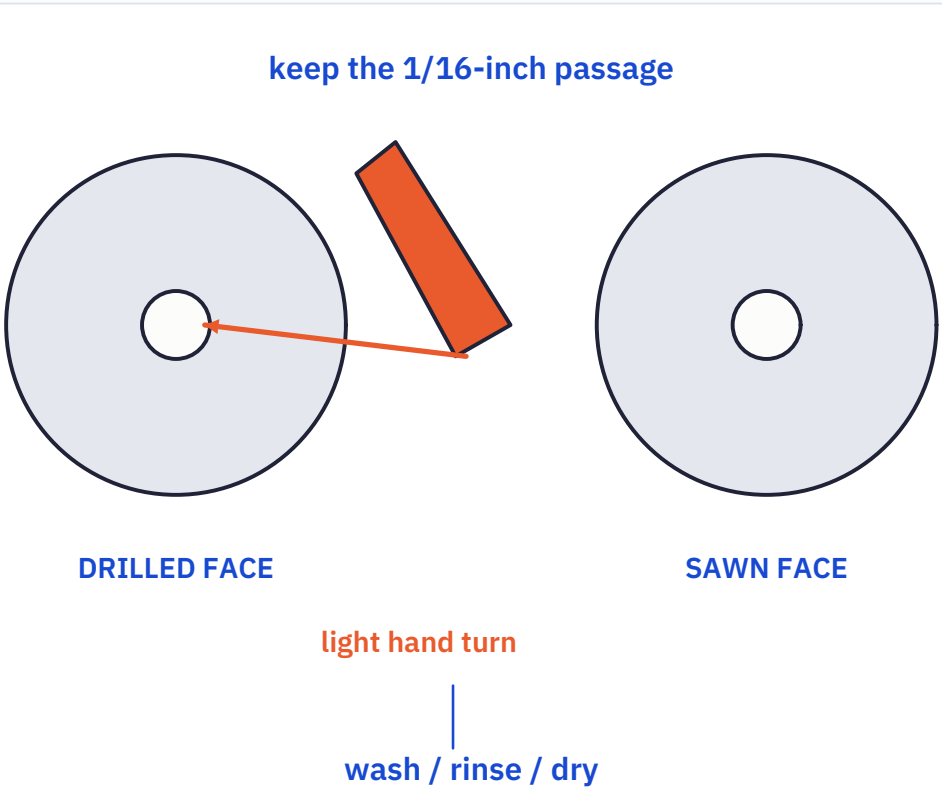
1. Set a square cut with the **24 TPI M42 blade at low speed**. Set the nominal slice to **2 mm ±0.2 mm**; account for the stop's blade reference and kerf.
2. Clamp the retained rod securely. Let the cap slice fall clear into a tray; do not hold the 2 mm slice by hand.
3. Measure thickness and squareness. Reject a wedge or damaged slice. Clean and reseat the remaining blank, then realign it for the next drilled hole.

NO FILLER HIDES A BAD SLICE

The cap must sit flat on the elbow. A wedge-shaped cut opens a weld gap and changes the passage length; it is rejected at this cutting step.

Clean both jet faces

Both ends of the bore are accessible now, before welding.



Inspect under magnification. A small edge burr is removed by hand, with the passage kept consistent.

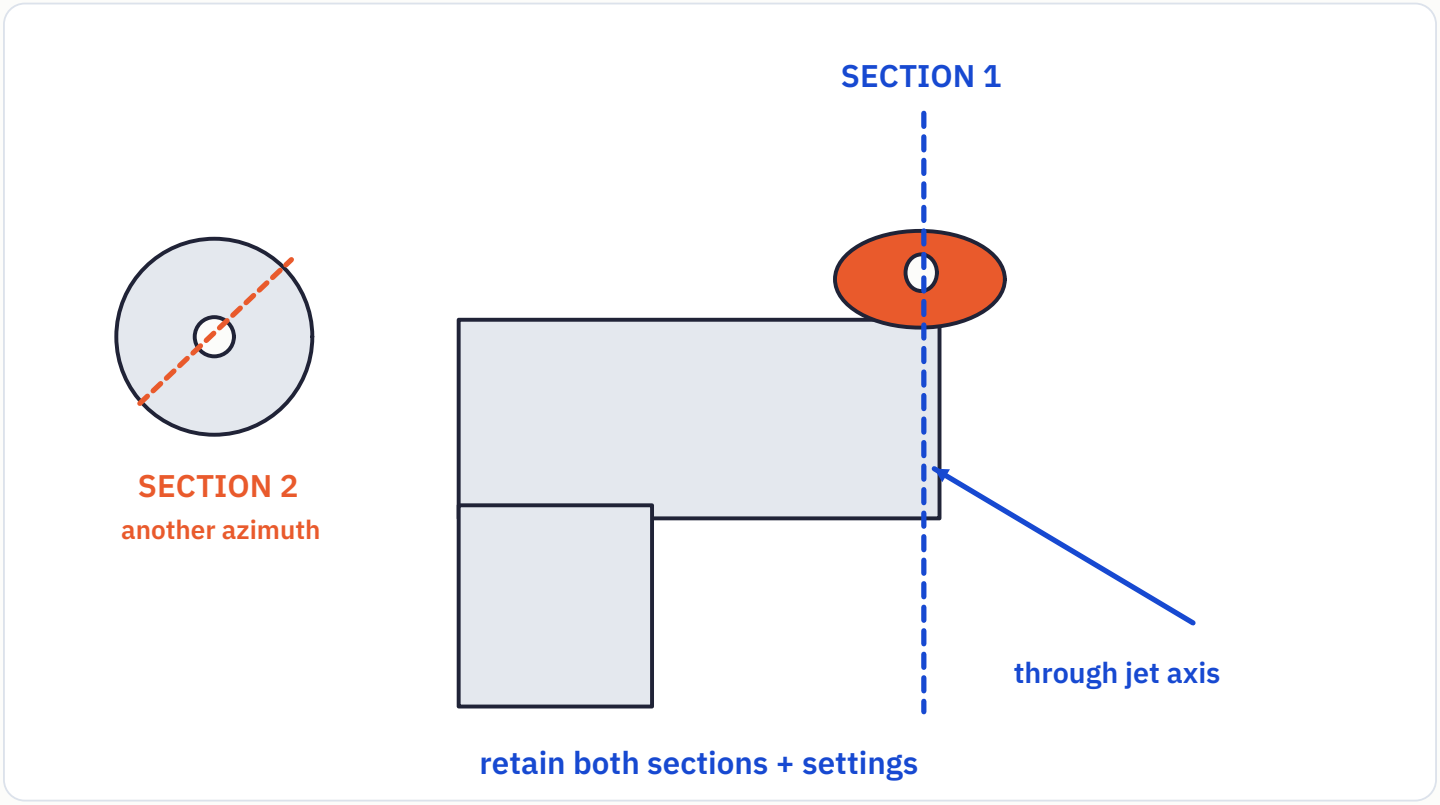
1. Inspect the drilled and sawn faces. Remove loose saw/drill burrs with a **light hand turn** of a larger owned drill if needed.
2. Keep the 1/16-inch passage and exit edge intact. Do not power-countersink this thin cap or turn a burr cleanup into a larger jet.
3. Wash away cutting fluid and loose particles, rinse and dry. Check that the cap sits on the measured elbow land without rocking.

LOOK THROUGH, THEN FIT

Reject a visibly distorted or obstructed hole. A post-weld drill-through repair changes this geometry and requires its own recorded acceptance.

Section the welded trial

An outside bead cannot show whether the bore-side root fused.



One cut passes through the jet axis. A second azimuth samples a tack or restart region.

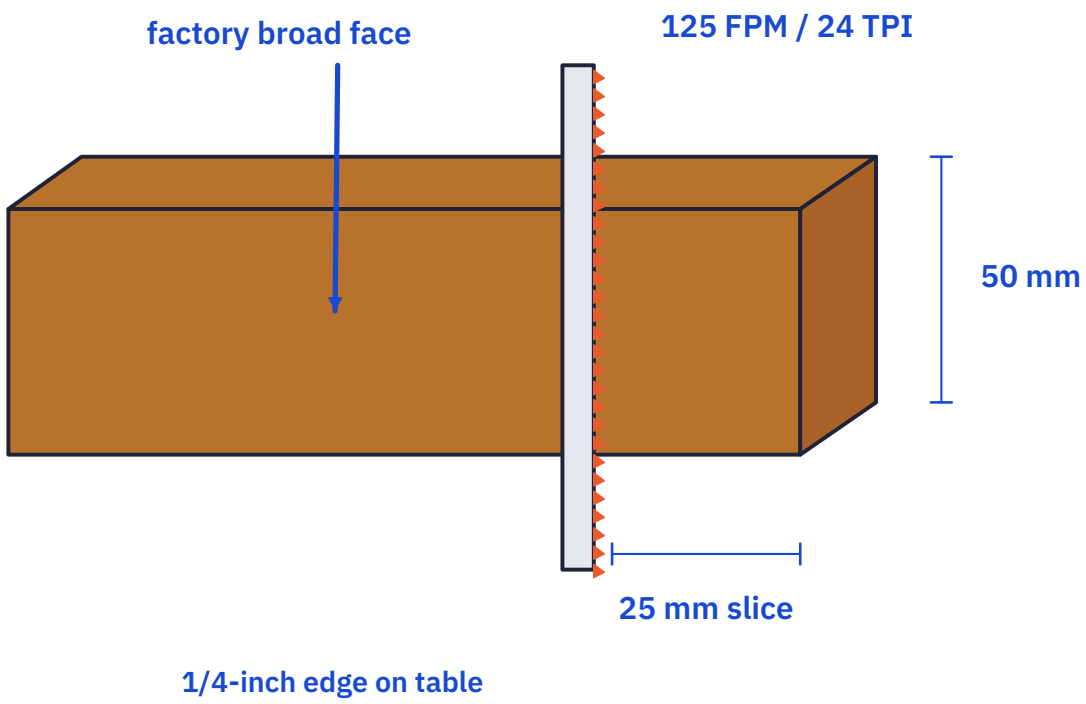
1. Secure the spare elbow body in the bandsaw vise. Section the trial cap/elbow **through the jet axis**, retaining the cut pieces.
2. Make the second section at a different azimuth. Dress the cut faces on a flat support using owned fine abrasive stock, then inspect and photograph under magnification.
3. Keep sections with the fit and weld settings record. Reject a bore-connected unfused interface, cracking, root scale or distorted passage.

SECTION READABILITY IS REQUIRED

A root that cannot be read remains unresolved. Sectioning samples the joint; appearance, pressure retention or a good jet stream do not prove full root fusion.

Cut the rotator contact shoe

One crosscut from the acquired nominal 1/4 × 2 inch C110 bar.



Stand the stock's 2-inch mill width vertically. The factory broad face contacts the tube; no copper hole or thread is made.

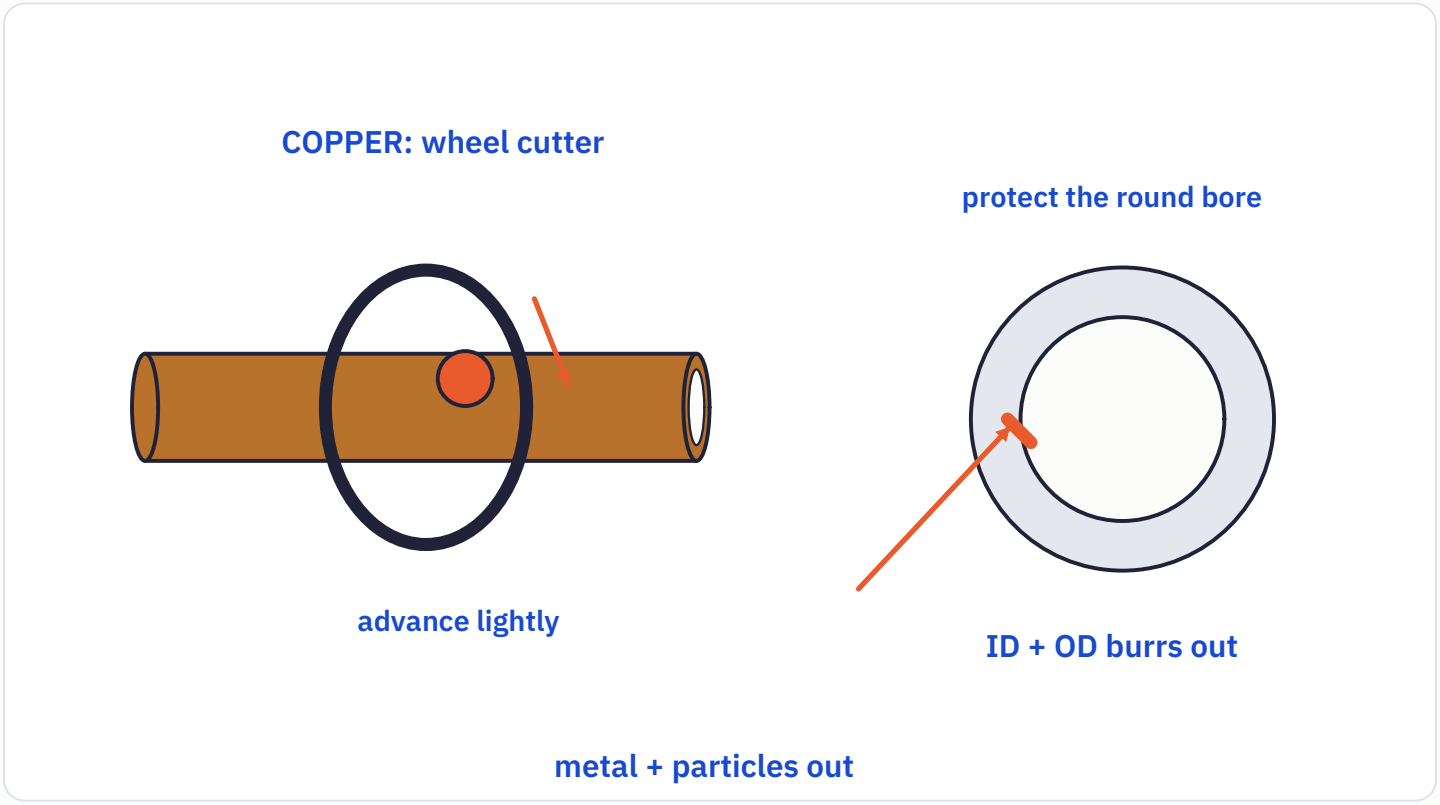
1. Put the **1/4-inch edge on the BA4555 table**, broad face vertical. Clamp the retained bar and mark one **25 mm slice**.
2. Use the installed **24 TPI M42 blade at 125 FPM** with light downfeed, letting the gullets clear. Keep the cutoff free.
3. Deburr the fresh cut without rounding the factory contact face. Stand the 50 mm stock width vertically in the ground-shoe shelf.

THE MILL FACE IS THE CONTACT

The shoe is nominally 6 × 25 × 50 mm. The saw cut seats on its printed shelf; it is not the sliding electrical contact surface.

Keep metal tube cuts clean

Square ends and clear bores matter on copper refrigerant lines and the stainless purge links.



Copper: RIDGID Model 150 wheel cutter. Stainless purge links: cut to the actual bench arrangement, then deburr and clean.

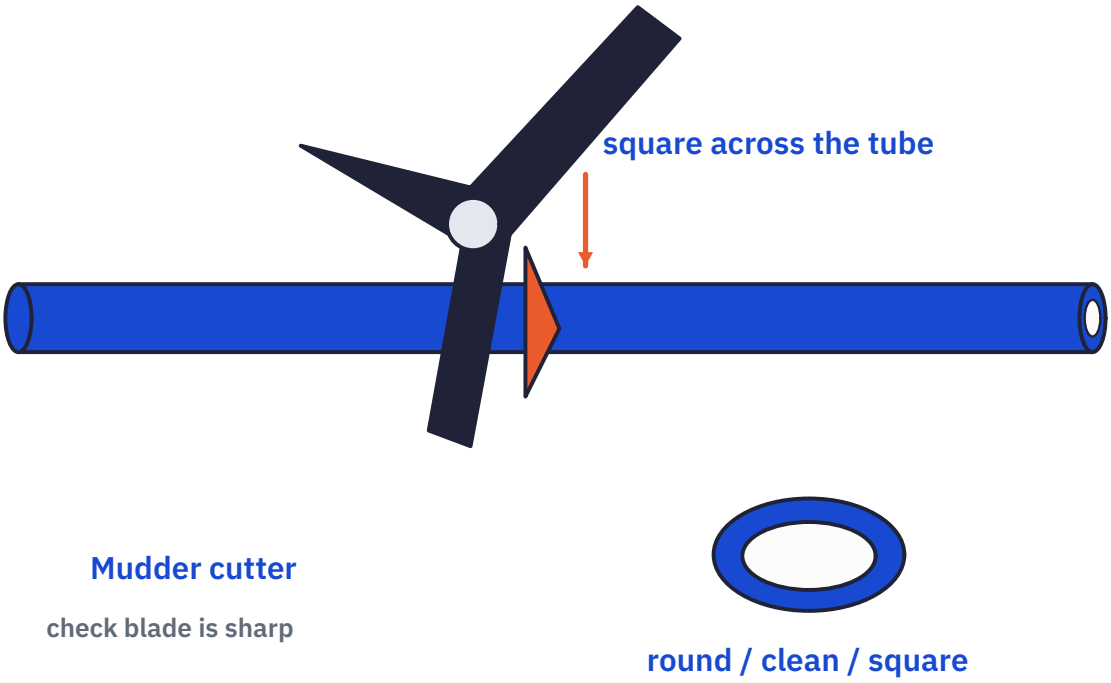
1. Measure the needed run and fitting engagements before cutting. Follow the **refrigeration guide** for coil stock, donor opening and capillary cuts.
2. Cut copper square with light cutter advances; remove ID/OD burrs and every particle without gouging or narrowing the bore. Protect open ends from debris.
3. For the jet purge fixture, cut two clean **1/4-inch OD stainless links** to suit the bench. Close the cylinder and depressurize before cutting the 6 mm regulated hose square.

CHARGED LINES HAVE THEIR OWN PROCEDURE

Use the refrigeration book before opening a donor. Capillary tubing takes its dedicated cutter. Fit stainless compression links to their maker's assembly instructions.

Square-cut the plastic tube

One clean cutting plane, with the bore round and the sealing surface undamaged.



Mudder cutter for 1/4-inch and 3/8-inch OD LLDPE. The printed razor miter box is a separate, untested tool trial.

1. Check the blade. Measure and mark the cut; place the tube square in the Mudder cutter without squeezing it out of round.
2. Make a clean cut across the tube. Inspect the end for angle, deformation and burrs; recut a damaged end before it reaches a fitting.
3. Allow the real fitting's full insertion at each end, mark engagement, push fully home and tug-test. Keep the smooth outside sealing surface free of scores.

END CONDITION BEFORE LENGTH

A nominally correct length with a slanted, oval or scored end is not ready for a push-fit joint. Use the received fitting's insertion requirement.

Cut the faucet tube set

Choose Black or White before pulling the flavor tubes from their reels.

BOTH FINISHES

blue / 1/4-inch	1540 mm
soda faucet / 3/8-inch	330 mm

BLACK FAUCET

2 flavor tubes	2 × 1963

WHITE FAUCET / FLAVOR

A	556 white + 1398 black
B	514 white + 1440 black

Lengths in mm. White faucet flavor runs join white upper tubes to black lower tubes using the staggered PP0408W unions.

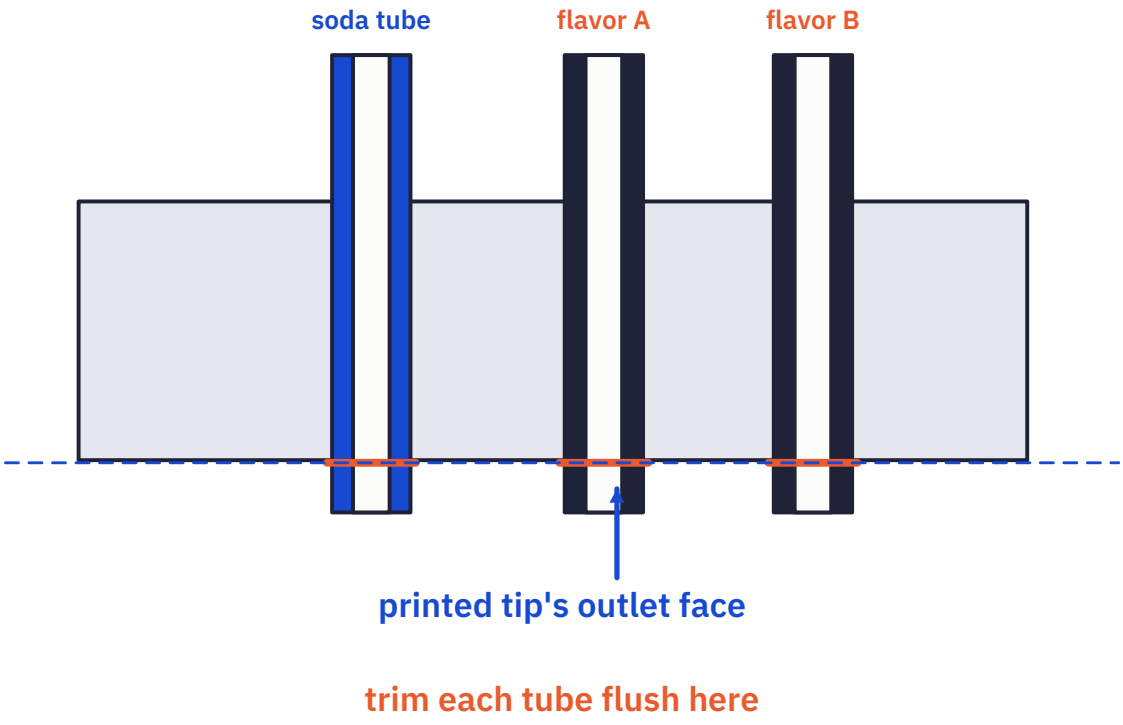
1. Cut the **blue 1/4-inch umbilical** and the finish's **3/8-inch soda faucet tube** to the common lengths shown.
2. For Black, cut two full black flavor tubes. For White, cut the separate A/B white and black pairs shown; keep the two union stations identified.
3. Square-cut every end. Route to the actual faucet and inspect the tails before the outlet trim; the blue tube begins at the Westbrass's lower compression port.

REACH IS PART OF THE CUT

These factory cuts include installation reach allowance. Do not substitute the shorter nominal installed lengths or equalize White faucet A/B pieces.

Trim the three outlets flush

The printed tip supports the tubes. The tubes remain the fluid passages.



Section schematic: one soda outlet and two flavor outlets share the printed tip's dispense-face plane.

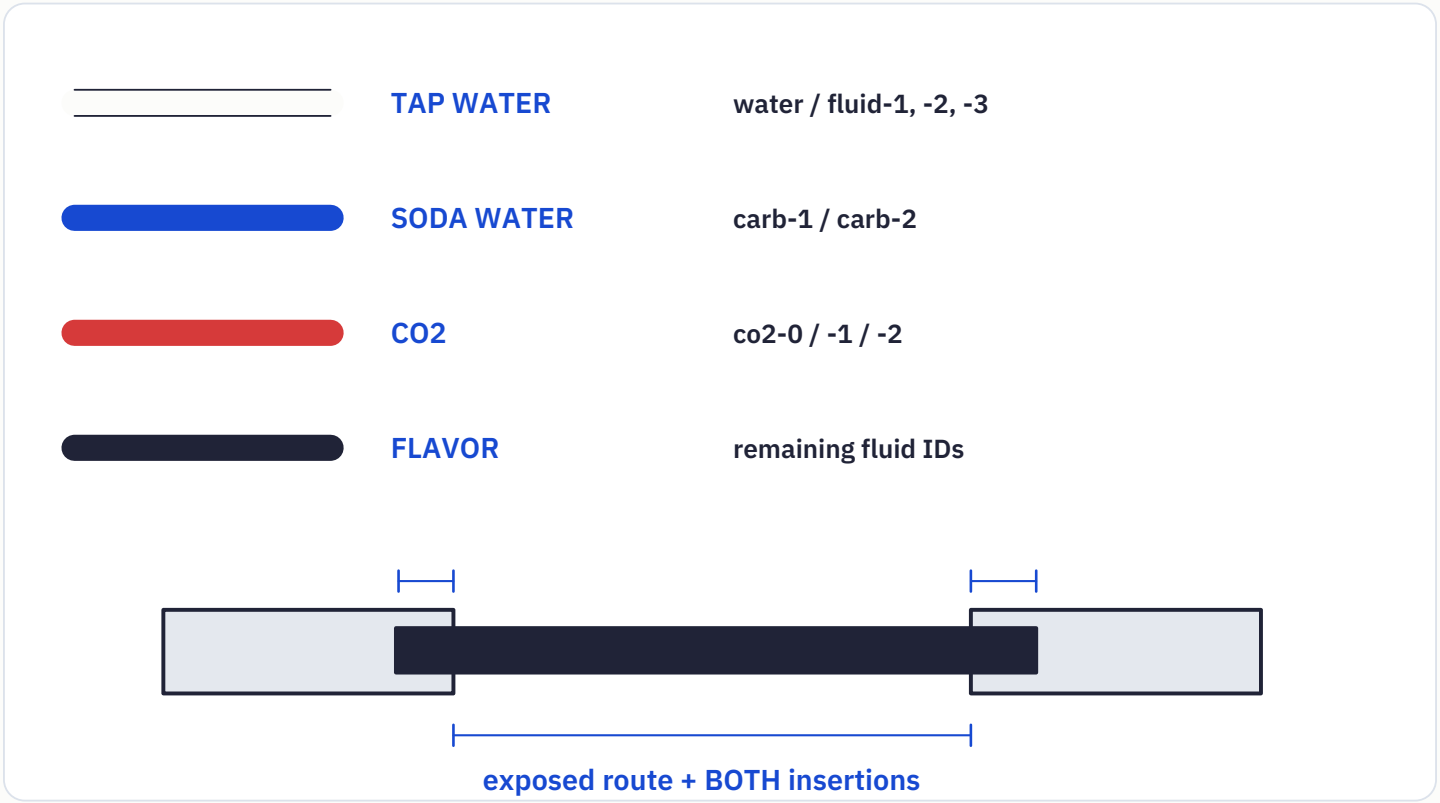
1. Route the 3/8-inch soda tube and both flavor tubes through their actual faucet channels. Fit the tip and verify that all three tubes pass freely.
2. Use a clean square cut to trim **each outlet flush with the printed tip**. Keep the blade away from the display, ribbon and printed sealing/support faces.
3. Inspect each outlet for an open round bore and a clean end. Check that the tubes and ribbon clear the seated display underside.

TRIM AT THE OUTLET PLANE

The factory tail allowance is retained. This is the three outlet-end trim; the lower bare tails still need their full installed reach and square push-fit ends.

Cut internal tubes by route

A developed exposed path is not a blank length until both engagements are included.



Use the current fluid topology and internal-plumbing procedure. Identify each segment before installation.

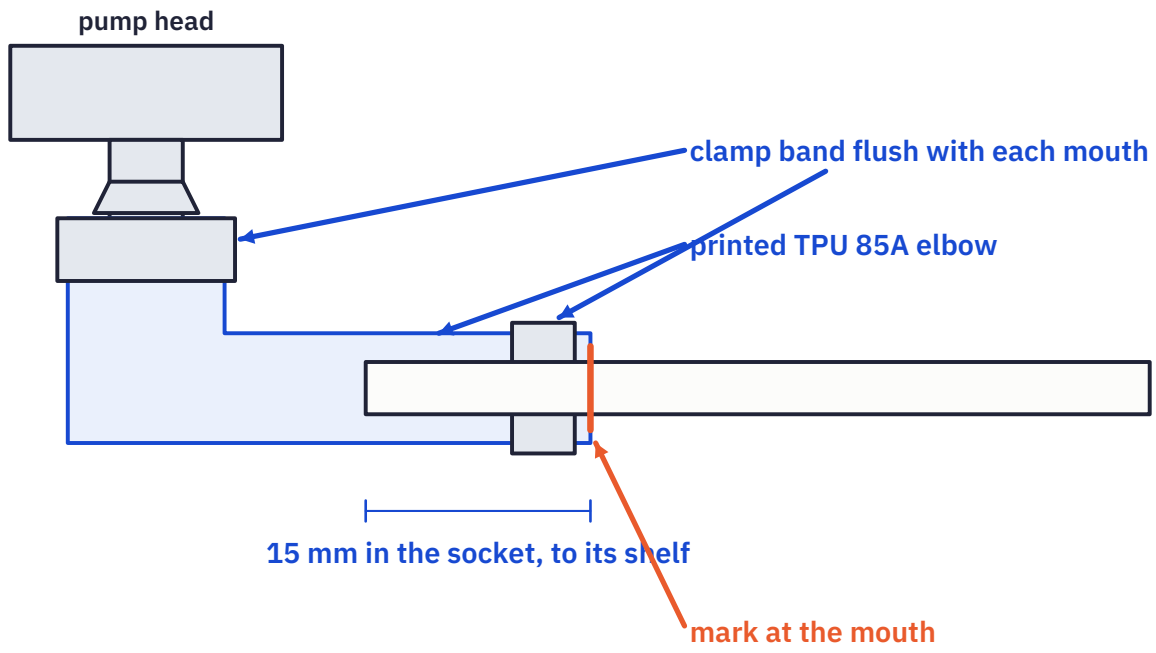
1. Pull from the carrying fluid's reel: **white tap water, blue soda water, red CO2, black flavor**. Fluid-1, -2 and -3 are white; tag all fluid IDs.
2. Cut established runs from their reported cut length. Bench-fit the four moving bowed tee links and the warm CO2 routes where fitting reaches and travel remain unqualified.
3. Include both received-fitting insertions even at zero exposed span. Keep the PRV vent mouth flush with the shell flank, its relief chase clear, and every finished end square.

NO FORCED ROUTE

Keep stock bend radii and collet axes. Resolve the documented fluid-2 clearance before assembly; a shorter cut is not a clearance repair. Tug-test each seated end.

Cut the pump elbow tubes

Each tube runs into a printed elbow's socket; the cut is the exposed route plus both grips.



water-7 ~70 mm / water-6 ~40 mm exposed, 1/4-inch white LLDPE

Water-7 suction and water-6 discharge are 1/4-inch white LLDPE into the printed TPU elbows on the G Ganen's barbs.

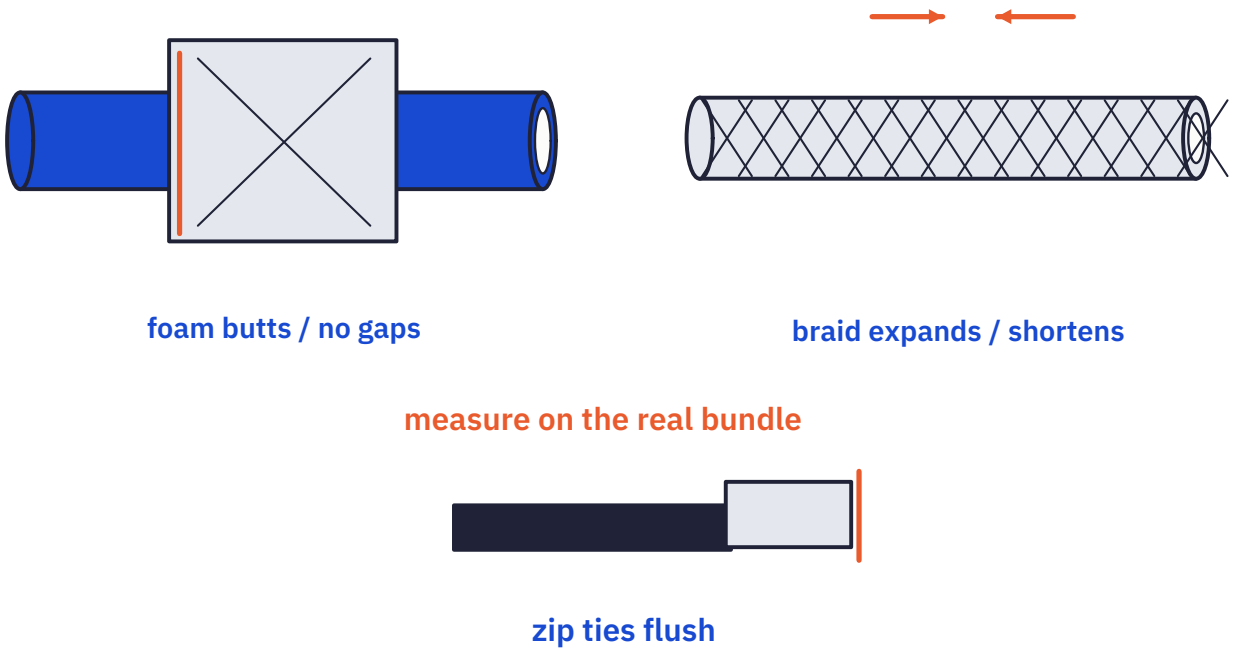
1. Add V-K's or the check chain's collet grip and the elbow's **15 mm** socket to each exposed route; the generated ~70 mm / ~40 mm paths are **not complete cuts**.
2. Square-cut and mark **15 mm** from the elbow end. Push in until the tube stops on the shelf with the mark at the mouth, then close the band flush with the mouth.
3. Square-cut the ASSE vent's 4 mm tube and mark its sleeve insertion. Cut the LLDPE funnel drain stub to **25.79 mm**.

FIT EACH SOCKET

The pump elbows take LLDPE in a printed socket under a clamp band; the ASSE vent's 4 mm tube takes its printed sleeve under ties. Neither end is a hose over a barb.

Cut coverings on the bundle

Foam butts tightly. Expanded braid takes up length, so measure it fitted.



The standard umbilical uses five foam/braid segments; internal soda risers use one continuous insulation piece per actual run.

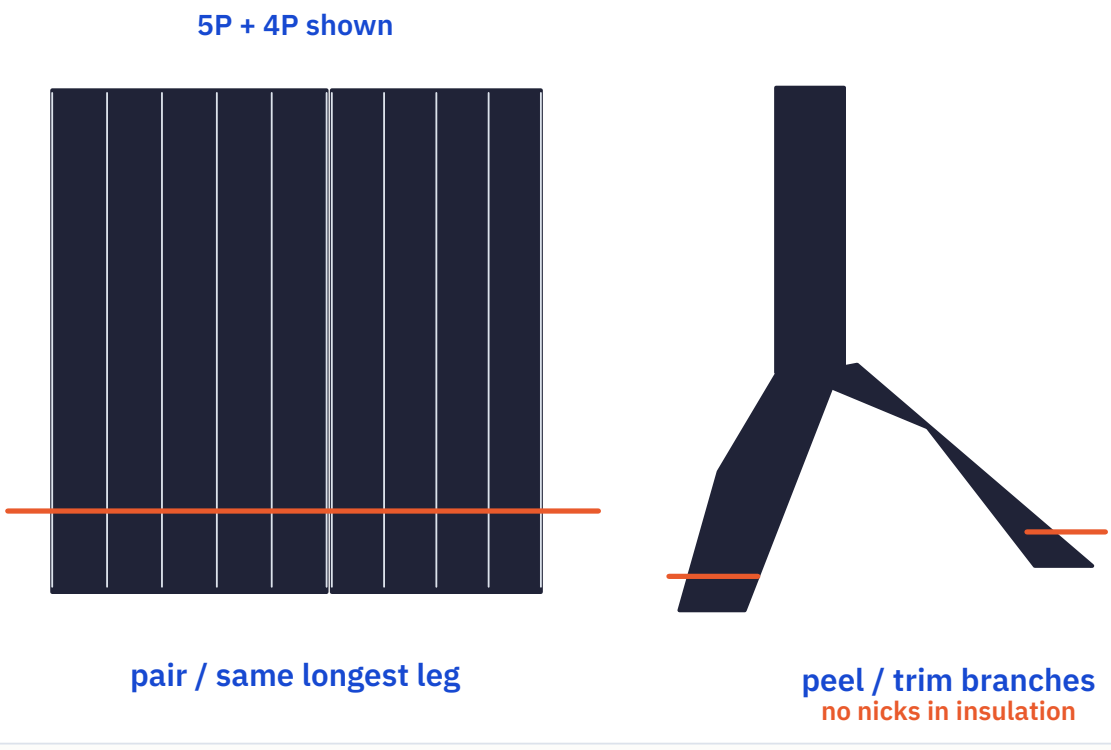
1. For the umbilical, fit the blue tube's foam over the prescribed **1338 mm** coverage with the **127 mm top** and **75 mm tail** left bare. Butt segments without gaps.
2. Measure each braid segment expanded over the actual tube/ribbon/foam pack before cutting; the top segment extends past its foam over the union stations.
3. Cut internal carb-1/carb-2 insulation to their complete runs before the collets close. Finish harness braid ends with heat-shrink, and flush-cut all zip-tie tails.

TRIM FOAM AFTER CURE IN THE MOLD GUIDE

Molded foam flash and silicone trimming belong with their cured parts. Keep knives off tubes, gasket seats and printed bearing faces when cutting any covering.

Cut the harness stock

Make the loom blank to its longest leg, then peel and trim the shorter branches.



Use the current cable-assembly and wiring run tables. A ribbon pair is two ribbons, laid edge-to-edge, cut to the same blank length.

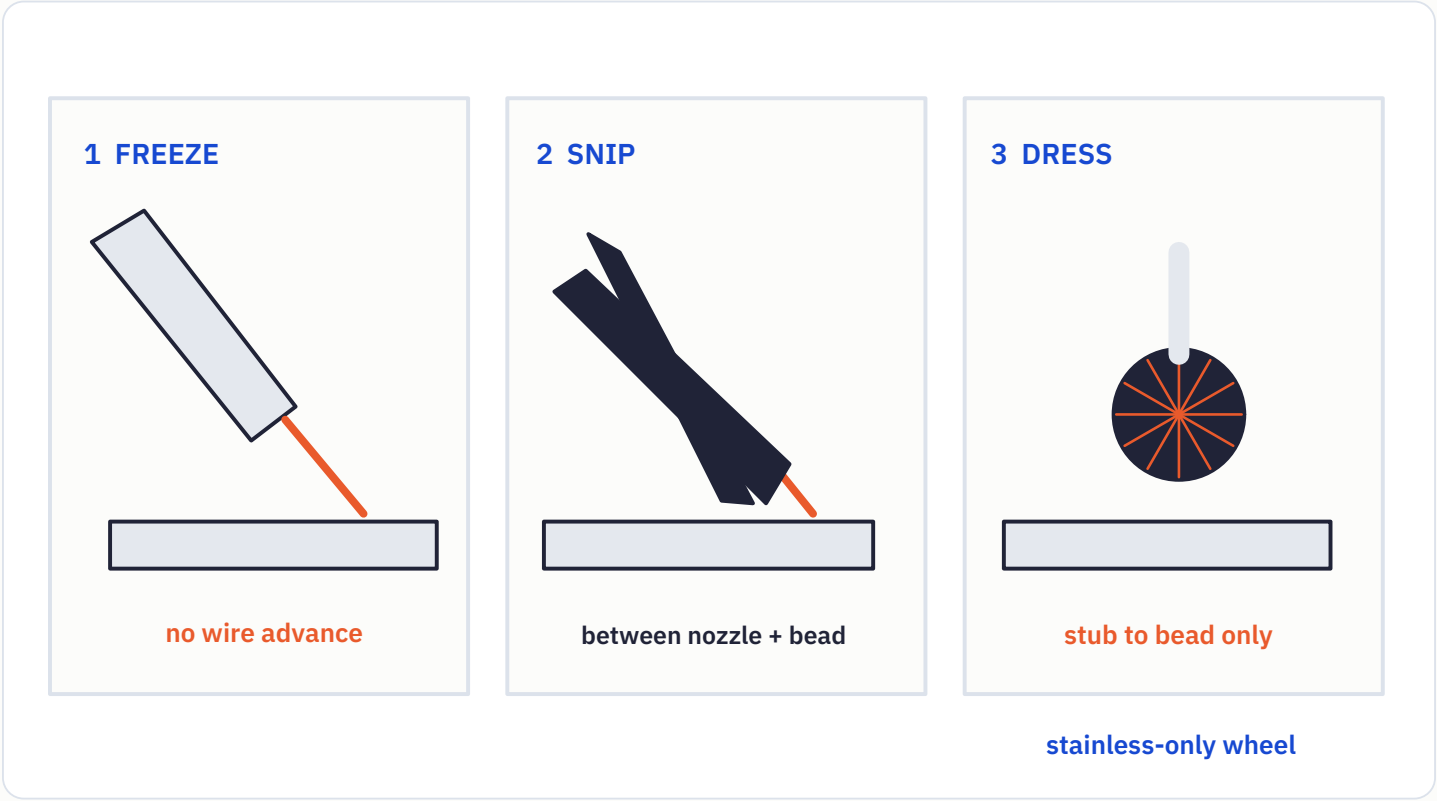
1. Choose the wafer's specified **22 AWG 3P/4P/5P ribbon pair**. Measure the longest leg plus the actual service loop, then cut both ribbons together.
2. Peel branch conductors without nicking insulation; trim the shorter legs to their endpoints. The 3P/5P web tear behavior still needs the stated stock check.
3. Use **16 AWG bulk** for the specified mains/trunk runs, **18 AWG SJOOW** for the compressor lead, and **28 AWG 4P** for the umbilical. Strip to the received terminal's barrel length.

LABEL THE LOOM BEFORE TERMINATION

DC-5 fixed half is 350 mm 22 AWG 4P; cartridge leads are four ~100 mm pieces. Keep the current donor's unverified external connector decision out of a speculative final cut.

Snip the attached filler wire

At the end of a bead, keep the gun where it stopped while the wire is detached.



One hand holds the stopped gun; the other brings the diagonal-cutter tips between wire nozzle and bead.

1. End the weld and hold the head at its stopping position. **Do not feed extra wire** from the feeder before detaching it.
2. Put the Knipex 70 11 110 tips between the nozzle and bead and snip the attached wire. Remove the freed head without lasering the stub off.
3. With the work supported and weld operation stopped, dress the stub with a stainless-only **1/4-inch-shank flap wheel in the drill**, down to the bead and no further.

PRESERVE THE CRATER FOR INSPECTION

The end-of-bead crater can contain a crack. Grinding into the bead can smear it shut before dye penetrant; do not grind a crater away to make the weld look finished.

Keep the source at hand

The current procedures own the dimensions and qualification records. The pictures give the working order.

End plates / vessel / rods 03-10 / 27

hardware/assembly/pressure-vessel.md

Jet fit, drilling, cutting and coupon 11-17 / 19

hardware/assembly/water-inlet-jet.md

Jet split-jaw fixture 13-14

hardware/printed-parts/fixtures/water-inlet-jet/README.md

Rotator contact shoe 18

hardware/printed-parts/fixtures/weld-rotator/README.md

Faucet / umbilical cut schedule 21-22 / 25

hardware/assembly/faucet-and-umbilical.md

Internal fluid routes + fit limits 23-24

hardware/assembly/internal-plumbing.md

Harness stock + branch schedule 26

hardware/assembly/cable-assemblies.md

Current wiring run table 26

hardware/wiring/ac-wiring-schedule.md

Companion books

Refrigeration: coil, donor opening, capillary and line work

Molds: pouring, cure and trimming foam / silicone

Weld rotator: fixture assembly and commissioning

Print on Letter, one-up, 100% / no scaling. The body is already 98% centered; the broad edge bands are drawn separately for the calibrated Epson borderless output.